C7 Oxidation, Reduction and Redox – Knowledge Organiser

AQA A-level Chemistry • Set 1: Definitions and Oxidation State Rules **Paper 1**

#	Question	Answer
1	Oxidation	Loss of electrons
2	Reduction	Gain of electrons
3	Oxidising agent	Electron acceptor
4	Reducing agent	Electron donor
5	Oxidation state of an uncombined element	0
6	Sum of oxidation states in a compound	0
7	Sum of oxidation states in an ion	Charge on the ion
8	Usual oxidation state of O	−2
9	Oxidation state of O in peroxides	−1
10	Oxidation state of O in OF_2	+2

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AQA A-level Chemistry • Set 2: More Oxidation State Rules **Paper 1**

#	Question	Answer
1	Usual oxidation state of H	+1
2	Oxidation state of H in metal hydrides	−1
3	Oxidation state of F in compounds	−1
4	Oxidation state of Group 1 metals	+1
5	Oxidation state of Group 2 metals	+2
6	Usual oxidation state of Cl in compounds	−1
7	Oxidation state of S in SO_4^{2-}	+6
8	Oxidation state of Mn in MnO_4^-	+7
9	Oxidation state of Cr in $Cr_2O_7^{2-}$	+6
10	Oxidation state of N in NO_3^-	+5

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AQA A-level Chemistry • Set 3: Half-Equations and Oxidation States **Paper 1**

#	Question	Answer
1	Half-equation	Equation showing electrons gained or lost by one species
2	Disproportionation	Same element oxidised and reduced
3	Half-equation for Fe^{2+} to Fe^{3+}	$Fe^{2+} \rightarrow Fe^{3+} + e^-$
4	Half-equation for Cl_2 to Cl^-	$Cl_2 + 2e^- \rightarrow 2Cl^-$
5	Oxidation state of Cl in ClO^-	+1
6	Oxidation state of Cl in ClO_3^-	+5
7	Oxidation state of N in NH_3	−3
8	Oxidation state of C in CO_2	+4
9	Oxidation state of Fe in Fe_2O_3	+3
10	Oxidation state of S in H_2S	−2